

A Novel Uplink Cognitive Rate-Splitting Scheme for UAV Communications

Dinh Trieu Duong, Manh Le-Tran, and Trinh Anh Vu

Abstract—This letter proposes a novel uplink cognitive communication for unmanned aerial vehicles (UAVs) that share the licensed spectrum of a primary user (PU). Motivated by spectrum scarcity in crowded unlicensed bands, we adopt a cognitive radio inspired rate-splitting multiple access (CR-RSMA) approach to protect the PU while supporting multiple UAV-based secondary users (SUs). In our proposed scheme, this approach provides a more general feasibility condition than conventional pairing schemes. For the scenario of equal-rate SUs, we derive the maximum admissible number of UAVs without violating the PU target rate. To realize this bound, we develop an explicit successive interference cancellation (SIC) design with scalar power splitting and a fixed decoding order, and further consider an RSMA design with private-rate guarantees. The framework is extended to unequal UAV channel gains without power control via a greedy strategy, and to multi-PU scenarios through a grouping algorithm. Simulation results confirm that the proposed RSMA-based scheme can support a higher number of UAVs than conventional scheme while strictly preserving the PU rate.

Index Terms—Cognitive radio, UAV communications, rate-splitting multiple access.

I. INTRODUCTION

Recently, unmanned aerial vehicles (UAV)-enabled wireless communications have gained much popularity for applications due to their flexible deployment and high mobility. Within 6G infrastructures, UAVs can work as a bridge to connect air space, ground, and underwater networks, which offers flexibility and enhanced connectivity in diverse environments [1]. Traditionally, UAVs operate on unlicensed bands. However, with recent advances in technology, many new wireless devices have been developed, which also operate on unlicensed bands. Therefore, these bands have become overcrowded and UAVs may face the problem of spectrum scarcity.

Cognitive radio (CR) has emerged as a promising solution for solving spectrum challenges. Specifically, UAVs are allowed to access the licensed or unlicensed spectrum to dynamically share their spectrum with the licensed users. To further improve spectral efficiency, CR-inspired rate-splitting multiple access (CR-RSMA) leverages dynamic spectrum access and the capacity benefits of RSMA [2]. Specifically, a secondary user (SU) splits its message so that part of the interference is decoded and the rest is treated as noise, improving spectrum utilization while meeting primary user (PU) constraints [3].

However, most existing works have mainly focused on evaluating performance of RSMA, without providing generalized conditions and optimal rate-splitting schemes for pairing a

PU with one or multiple SUs. For CR-NOMA, [4] and [5] proposed user pairing techniques to improve overall network throughput, but these techniques are not directly applicable to CR-RSMA, where partial interference decoding and more flexible splitting structures fundamentally change feasibility and optimality. For uplink RSMA, [6] studied user pairing and resource allocation, yet key issues for an uplink CR-RSMA framework, such as generalized feasibility conditions for pairing a PU with multiple SUs and an explicit rate-splitting design that preserves the PU rate, remain under-explored. In prior work [7], authors proposed a PU-SU pairing condition, but it was limited to the single PU-SU case.

In this letter, we propose an uplink cognitive RSMA (CRS) system that enables multiple UAV-based SUs to share the spectrum of a licensed PU while preserving the PU target rate. Our proposed CRS scheme first leverages the Gaussian multiple access channel (MAC) [8] capacity region to characterize the feasibility of the equal-rate tuple and to obtain the maximum admissible number of UAVs under an equal-rate scenario. Then, we propose a design principle with power splitting at each UAV and a fixed decoding order at the base station (BS) to realize the target PU-SU rates. We further propose an optimization framework with private-rate guarantees, and a greedy admission strategy for scenario without power control, which also extends to multi-PU scenario via simple grouping. Results confirm that our proposed CRS scheme admits more UAVs than conventional schemes while strictly maintaining the PU rate. Also, the proposed optimization shows significant gains over the conventional schemes.

II. SYSTEM MODEL

Consider the uplink of a single cell where a BS receives from one PU and a set of U potential UAV-based SUs. All terminals are single-antenna and operate in the same frequency band. The received signal at the BS is

$$y = h_p \sqrt{P_p} x_p + \sum_{i=1}^U h_{s,i} \sqrt{P_{s,i}} x_{s,i} + n, \quad (1)$$

where x_p and $x_{s,i}$ are unit-power information symbols of the PU and the i -th UAV, respectively, h_p and $h_{s,i}$ are complex channel gains, P_p and $P_{s,i}$ are transmit powers, and $n \sim \mathcal{CN}(0, 1)$ is normalized additive white Gaussian noise. The BS is assumed to have perfect channel state information, while the PU and UAVs know only their target rates.

For simplicity, define the received signal-to-noise ratios $\gamma_p = P_p |h_p|^2$ and $\gamma_{s,i} = P_{s,i} |h_{s,i}|^2$ while the PU requires a fixed target rate $R_p > 0$. For the power control case, each UAV must achieve the same rate $R_s > 0$. With a set $\mathcal{N} \subseteq \{1, \dots, U\}$ of

The authors are with the Faculty of Electronics and Telecommunications, University of Engineering and Technology, Vietnam National University, Hanoi 123106, Vietnam.

In the proposed C-RS,

N active UAVs, the uplink behaves as a $(N+1)$ -users MAC, where the PU has received SNR γ_p with rate R_p , and each UAV $i \in \mathcal{N}$ has received SNR $\gamma_{s,i}$ with rate R_s .

The fundamental rate limitations of this uplink multiuser channel are described by the capacity region of the Gaussian MAC [8]. For CRS, we first determine, for given (R_p, R_s) and channel gains, the maximum admissible number of UAVs $N_{\max}$ such that the associated rate tuple lies inside the uplink MAC capacity region. We then consider a decoding structure at the BS that attains such admissible tuples, thereby turning this theoretical bound into a practical receiver architecture.

Without loss of generality, we assume that

III. CRS WITH POWER CONTROL

We focus on scenarios where multiple UAV links generate similar uplink traffic and are modeled with an equal SU target rate R_s . Therefore, all admitted UAVs adjust their transmit powers so that the BS observes the same received SNR γ_s for all $i \in \mathcal{N}$. This equal-gain model is consistent with standard uplink power control and makes admission depend only on how many secondary links can be supported without violating the PU operation. Consequently, for a given set $\mathcal{N}$ of admitted UAVs, the rate tuple $(R_p, R_s, \dots, R_s)$ must lie within this capacity region for $\mathcal{N}$ to be admissible.

A. Maximum number of UAVs from MAC constraints

In this work, based on the uplink MAC capacity region from Section II, we derive $N_{\max}$ as the maximum number of admitted UAVs while preserving the PU's rate. Specifically, the single-user constraint for the PU is

$$R_p \leq \log_2(1 + \gamma_p). \quad (2)$$

For any fixed set $\mathcal{N}$ of N admitted UAVs with equal received SNR γ_s , the uplink is a $(N+1)$ -user Gaussian MAC. For the subset containing the PU and any m out of the N UAVs, the aggregate received SNR is $\gamma_p + m\gamma_s$, while the target sum-rate is $R_p + mR_s$. The MAC sum-rate bound [8] then imposes

$$R_p + mR_s \leq \log_2(1 + \gamma_p + m\gamma_s), \quad m \in \{1, \dots, N\}, \quad (3)$$

which is the key feasibility constraint relating R_p , the UAV rate R_s , and the number of admitted UAVs.

It is convenient to rewrite (3) as $R_s \leq f(m)$, where

$$f(x) = \frac{\log_2(1 + \gamma_p + x\gamma_s) - R_p}{x}, \quad x > 0. \quad (4)$$

Lemma 1: Assume $\gamma_p > 0$, $\gamma_s > 0$, and (2) holds. Then it is worth noting that $f(x)$ in (4) is strictly decreasing on $(0, \infty)$. Proof: See Appendix A. Lemma 1 implies that, as m grows, the right-hand side of (3) becomes smaller, so the tightest sum-rate constraint corresponds to the largest subset size.

Proposition 1: Suppose $\gamma_p, \gamma_s > 0$, and (2) holds. For a given $R_s > 0$, let $x^* > 0$ be the unique solution of $R_s = f(x)$. Then the maximum admissible number of UAVs with equal rate R_s is $N_{\max} = \lfloor x^* \rfloor$, where $\lfloor \cdot \rfloor$ denotes the floor operator. Proof: See Appendix B.

Since $f(x)$ is strictly decreasing, the equation $R_s = f(x)$ has a unique real solution x^* . Also, x^* can be found numerically

using a simple one-dimensional search, such as the bisection algorithm on an interval that brackets the solution.

Remark 1: If $N_{\max} = 0$, then even a single UAV cannot share the band with the PU. When $N_{\max} = 1$, we have the classical cognitive condition $R_p + R_s \leq \log_2(1 + \gamma_p + \gamma_s)$.

B. CRS structure and SIC order

We now construct an SIC structure that achieves the equal SU rate condition for $N_{\max}$ under the predefined rate of PU. Unlike conventional approaches, where the decoding strategy must be redesigned for each UAV link realization, our proposed CRS leverages standard uplink power control to enforce an equal-gain model at the BS. Consequently, the receiver can adopt a fixed SIC order and determine the splitting parameters $\{\delta_i\}$ via a simple construction that depends only on γ_s and the PU constraint, which is directly implementable for dense UAV deployments. More specifically, each i -th UAV $i \in \{1, \dots, N\}$ allocates its transmit power such that its effective received SNR can be decomposed into two components as

$$\gamma_s = (\gamma_s - \delta_i) + \delta_i, \quad 0 < \delta_i < \gamma_s. \quad (5)$$

Denote the power-splitting coefficients as $\delta = (\delta_1, \dots, \delta_N)$. Each UAV then transmits two independent streams $U_{i,1}$ and $U_{i,2}$ with effective received SNRs $\gamma_s - \delta_i$ and δ_i , respectively. Here, $U_{i,2}$ is a common-decoded sub-stream at the BS, while $U_{i,1}$ is the private sub-stream of UAV i . The PU transmits a single stream with effective received SNR γ_p , so there are $2N+1$ virtual streams $U_{1,1}, \dots, U_{N,1}, P, U_{N,2}, \dots, U_{1,2}$ decoded at the BS in this fixed SIC order. Under unit-variance noise, define the achievable rate for decoding a stream with signal power S with aggregate interference power I as

$$C(S, I) = \log_2(1 + S/(1 + I)). \quad (6)$$

Let $D_k = \sum_{j=1}^k \delta_j$ with $D_0 = 1$. The PU is decoded in the presence of the overall interference D_N , hence

$$R_{PU}(\delta) = C(\gamma_p, D_N). \quad (7)$$

For i -th UAV, the total rate under the same SIC order is the sum rate of the private part $U_{i,1}$ and the common part $U_{i,2}$ as

$$\begin{aligned} R_i(\delta) &= R_{i1}(\delta) + R_{i2}(\delta) \\ &= C(\gamma_s - \delta_i, (N-i)\gamma_s + \gamma_p + D_N - \delta_i) + C(\delta_i, D_{i-1}). \end{aligned} \quad (8)$$

The next step is to show that one can choose $\delta_1, \dots, \delta_N$ such that $R_i = R_s$ for all i and $R_{PU} = R_p$.

Lemma 2: For any (N, γ_p, γ_s) and $\delta_1, \dots, \delta_N \in (0, \gamma_s)$, the function $\phi_i(\delta_i) = R_i(\delta_i)$ defined on $(0, \gamma_s)$ is continuous and strictly increasing in δ_i .

This follows from Gaussian rate formulas that any shift of SNR from the heavily interfered part to the almost interference-free part strictly increases the total rate. Assume that

$$R_p + NR_s = \log_2(1 + \gamma_p + N\gamma_s). \quad (9)$$

It is worth noticing that

Remark 2: With the proposed CRS framework, there exist $\delta_1, \dots, \delta_N \in (0, \gamma_s)$ such that $R_i(\delta) = R_s, i = 1, \dots, N$ and $R_{PU}(\delta) = R_p$. The reason is that SIC with Gaussian codebooks can achieve any rate point on the uplink MAC

Algorithm 1 Recursive Construction of δ for Equal UAV Rates**Require:** $N, \gamma_p, \gamma_s, R_p, R_s$, tolerance $\varepsilon > 0$ **Ensure:** $\delta = (\delta_1, \dots, \delta_N)$

- 1: Find D_N^* such that $C(\gamma_p, D_N^*) = R_p$ by bisection over $D_N \in (0, N\gamma_s)$.
- 2: $D_0 \leftarrow 0$.
- 3: **for** $i = 1$ **to** $N - 1$ **do**
- 4: $\bar{\delta}_i \leftarrow \min\{\gamma_s, D_N^* - D_{i-1}\}$.
- 5: Solve $\delta_i \in (0, \bar{\delta}_i)$ such that $R_i(\delta) = R_s$ by bisection on δ_i , with $D_N = D_N^*$ and fixed D_{i-1} .
- 6: $D_i \leftarrow D_{i-1} + \delta_i$.
- 7: **end for**
- 8: $\delta_N \leftarrow D_N^* - D_{N-1}$.
- 9: **return** δ when $|R_N(\delta) - R_s| \leq \varepsilon$.

thus

sum-rate boundary, and our proposed scheme can provide sufficient degrees of freedom to realize the equal-rate tuple while meeting the PU rate constraint.

Since $R_i(\delta)$ depends on δ_i only through (δ_i, D_{i-1}, D_N) under the fixed SIC order, the power-splitting δ can be obtained by a recursive one-dimensional construction as summarized in the proposed **Algo. 1**.

Algorithm 1**C. Rate Optimization With Private-Rate Guarantees**

While **Proposition 1** specifies how many UAVs can be admitted for a given rate R_s , it does not specify how this rate is split between the two **RSMA streams**. In many UAV applications, however, a fixed portion of the rate must be carried on a more interference-robust private stream $U_{i,1}$. This is because the other stream $U_{i,2}$ is decoded under residual multiuser interference and is more vulnerable to imperfect SIC and channel-state uncertainty, which can degrade reliability in dense UAV uplinks. To capture this, we impose a minimum private-rate ratio and optimize the power splits accordingly.

Specifically, we introduce a design parameter $\alpha \in (0, 1)$ and require that each UAV achieves a minimum private-part rate such as $R_{i,1}(\delta) \geq \alpha R_s$. For a predetermined minimum rate $R_s^{\min}$, then we have the following optimization problem

$$(P0) \quad \max_{\delta, R_s} R_s, \quad (10a)$$

$$\text{s.t.} \quad R_{PU}(\delta) \geq R_p, \quad (10b)$$

$$R_i(\delta) \geq R_s, \quad i = 1, \dots, N, \quad (10c)$$

$$R_{i,1}(\delta) \geq \alpha R_s, \quad i = 1, \dots, N, \quad (10d)$$

$$R_s \geq R_s^{\min}, \quad (10e)$$

$$0 < \delta_i < \gamma_s, \quad i = 1, \dots, N. \quad (10f)$$

Here, (10b) enforces the PU rate requirement, (10c) guarantees a UAV rate, and (10d) ensures that each UAV achieves at least αR_s on its private stream. The problem is non-convex in δ since the rate expressions involve ratios of linear terms in δ_i . Therefore, we exploit a difference-of-convex representation and apply successive convex approximation (SCA) [9]. The signal and interference terms in (6) can be written as linear functions $S(\delta)$ and $I(\delta)$, so we have

$$C(S(\delta), I(\delta)) = \log_2(1 + S(\delta) + I(\delta)) - \log_2(1 + I(\delta)). \quad (11)$$

Algorithm 2 Private-Rate Guarantees Optimization**Require:** $\gamma_p, \gamma_s, R_p, R_s^{\min}, \alpha, N, I_{\max}$, tolerance $\varepsilon > 0$.**Ensure:** Power-splitting δ and UAV rate R_s .

- 1: Initialize $(\delta^{(0)}, R_s^{(0)})$ satisfying (10); set $t \leftarrow 0$.
- 2: **repeat**
- 3: Compute $D_k^{(t)} = \sum_{j=1}^k \delta_j^{(t)}$ for $k = 1, \dots, N$.
- 4: Compute all $I(\delta^{(t)})$ in R_{PU} , R_i , and $R_{i,1}$; construct $\tilde{R}_{PU}^{(t)}$, $\tilde{R}_i^{(t)}$, and $\tilde{R}_{i,1}^{(t)}$ using (11) and (12).
- 5: Solve the convex problem (P1) to obtain (δ^*, R_s^*) .
- 6: Update $\delta^{(t+1)} \leftarrow \delta^*$ and $R_s^{(t+1)} \leftarrow R_s^*$.
- 7: Set $t \leftarrow t + 1$.
- 8: **until** $|R_s^{(t)} - R_s^{(t-1)}| \leq \varepsilon$ or $t = I_{\max}$
- 9: **return** $\delta^{\text{SCA}} = \delta^{(t)}$, $R_s^{\text{SCA}} = R_s^{(t)}$.

Hence, $C(S(\delta), I(\delta))$ becomes a difference of two concave logarithms whose arguments are linear in δ . Let $g(x) = \log_2(1 + x)$, which is concave for $x \geq 0$. Around $x^{(t)} \geq 0$ we use upper bound

$$g(x) \leq g(x^{(t)}) + g'(x^{(t)})(x - x^{(t)}), \quad g'(x) = \frac{\log_2 e}{1 + x}. \quad (12)$$

Applying (12) to $\log_2(1 + I(\delta))$ at $I(\delta^{(t)})$ yields a lower bound

$$C(S(\delta), I(\delta)) \geq \tilde{C}^{(t)}(S(\delta), I(\delta)), \quad (13)$$

which is tight at $\delta = \delta^{(t)}$. Using this construction for R_{PU} , R_i , and $R_{i,1}$ gives $\tilde{R}_{PU}^{(t)}$, $\tilde{R}_i^{(t)}$, and $\tilde{R}_{i,1}^{(t)}$ satisfying $\tilde{R}_{PU}^{(t)} \leq R_{PU}$, $\tilde{R}_i^{(t)} \leq R_i$, and $\tilde{R}_{i,1}^{(t)} \leq R_{i,1}$, with equality at $\delta^{(t)}$.

At iteration t , we replace the rate constraints in (10) by these lower bounds and solve the equivalent convex problem

$$[\dots] (P1) \quad \max_{\delta, R_s} R_s, \quad (14a)$$

$$\text{s.t.} \quad \tilde{R}_{PU}^{(t)}(\delta) \geq R_p, \quad (14b)$$

$$\tilde{R}_i^{(t)}(\delta) \geq R_s, \quad i = 1, \dots, N, \quad (14c)$$

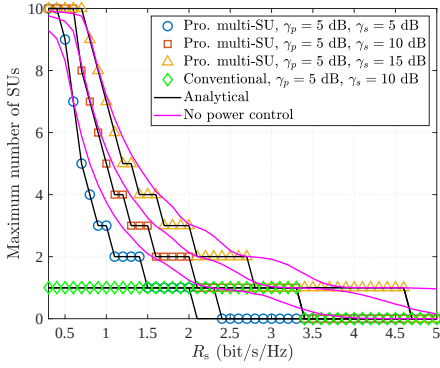
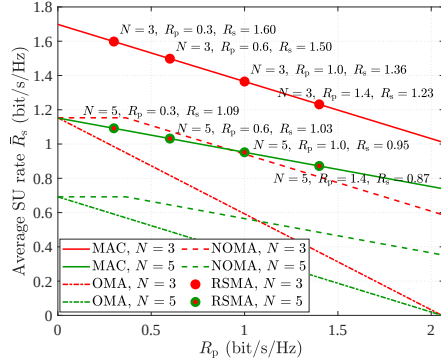
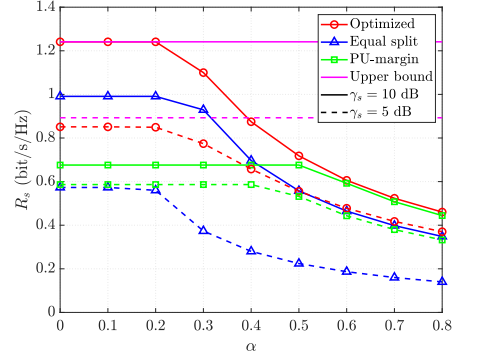
$$\tilde{R}_{i,1}^{(t)}(\delta) \geq \alpha R_s, \quad i = 1, \dots, N, \quad (14d)$$

$$(10e), (10f). \quad (14e)$$

that can be solved by standard convex solvers [9]. Let (δ^*, R_s^*) be its solution, we then set $\delta^{(t+1)} \leftarrow \delta^*$ and $R_s^{(t+1)} \leftarrow R_s^*$, and repeat until $|R_s^{(t+1)} - R_s^{(t)}| \leq \varepsilon$ or a maximum number of iterations is reached. In simulation, we use a simple step-size control to maintain feasibility of the original constraints in (10). After each update, we check the rate equations and limit R_s to the sum-rate upper bound. The process is presented in proposed **Algo. 2**. This process generates a non-decreasing sequence of R_s and converges to a stable point, while enforcing PU protection and a minimum private-rate fraction α .

IV. RSMA WITHOUT POWER CONTROL AND MULTI-PU

The equal-gain setting assumed that all admitted UAVs can be driven to the same received SNR levels at the BS via uplink power control. In many practical deployments, however, such power control is unavailable or undesirable due to UAV energy constraints, limited feedback, and fast channel fluctuations. As a result, UAVs typically transmit with a fixed power budget

Fig. 1. Maximum number of SUs versus R_s .Fig. 2. Average SU rate $\bar{R}_s$ versus PU rate R_p .Fig. 3. Optimized UAV rate R_s versus α .

Algorithm 3 Greedy Admission Without Power Control

Require: $R_p, R_s, \gamma_p, \{\gamma_{s,i}\}_{i=1}^U$
Ensure: $N_{np}, \{\gamma_{(j)}\}_{j=1}^{N_{np}}$

- 1: Sort $\{\gamma_{s,i}\}$ to get $\gamma_{(1)} \geq \dots \geq \gamma_{(U)}$
- 2: $k \leftarrow 0, Q_0 \leftarrow 0$
- 3: **while** $k < U$ **do**
- 4: **if** $R_p + (k+1)R_s \leq \log_2(1 + \gamma_p + Q_k + \gamma_{(k+1)})$ **then**
- 5: $k \leftarrow k + 1, Q_k \leftarrow Q_{k-1} + \gamma_{(k)}$
- 6: **else**
- 7: **break**
- 8: **end if**
- 9: **end while**
- 10: $N_{np} \leftarrow k$

and experience heterogeneous path-loss and fading, which leads to different received SNRs $\gamma_{s,i}$. This section therefore targets a deployment-oriented setting where admission must be decided directly from the instantaneous heterogeneous SNRs and extended to multiple PUs via a low-overhead grouping procedure. In this case, the PU constraint remains as in (2) while each candidate SU must at least satisfy its own condition

$$R_s \leq \log_2(1 + \gamma_{s,i}), \quad i = 1, \dots, U. \quad (15)$$

Then it could maintain the target rate if scheduled alone with the PU.

To exploit the difference of the UAV links, the BS orders the effective received SNRs in non-increasing order as $\gamma_{(1)} \geq \gamma_{(2)} \geq \dots \geq \gamma_{(U)}$ and progressively adds the strongest UAVs as long as the joint rate requirement of the PU and the already admitted UAVs remains inside the instantaneous uplink capacity region. For any $k \in \{0, 1, \dots, U\}$, define the cumulative received SNR of the k strongest UAVs as

$$Q_k = \sum_{j=1}^k \gamma_{(j)}, \quad Q_0 = 0. \quad (16)$$

If the first k UAVs are admitted together with the PU, the target sum-rate is $R_p + kR_s$, and the corresponding sum-capacity is $\log_2(1 + \gamma_p + Q_k)$. The condition for whether the $(k+1)$ -th UAV can be admitted is

$$R_p + (k+1)R_s \leq \log_2(1 + \gamma_p + Q_k + \gamma_{(k+1)}), \quad (17)$$

which directly leads to the following admission procedure.

For each channel realization, the proposed in [Algo. 3](#) returns an integer N_{np} that plays the same role as N_{max} in the equal-gain case, but it can be utilized in the heterogeneous received SNRs. Once the admitted set is fixed, each i -th UAV splits its received SNR $\gamma_{(i)}$ into two streams, and the BS decodes the resulting streams in the same fixed SIC order as in Section III.

In multi-PU systems, we can consider the greedy admission rule for each PU and perform a simple SNR-based grouping to form parallel PU-UAV clusters. This directly facilitates practical fixed-power UAVs such as $P_{s,i} = P_s$ with heterogeneous links $\gamma_{s,i} = P_s |h_{s,i}|^2$. After grouping, the RSMA structure and the SIC order in Section III-B are applied independently to each cluster. Consequently, this enables scalable dense UAV deployments while preserving all PU rate constraints.

V. NUMERICAL RESULTS

In this section, we present simulation results to verify and demonstrate the performance of the proposed uplink cognitive RSMA framework. We consider one PU and each point is averaged over 10^5 independent channel realizations. We study both the equal rate setting, where all admitted UAVs are observed at the BS with the same γ_s . In the no-power-control setting, UAV channels are modeled as Rayleigh fading.

In Fig. 1, we show the average maximum number of admitted SUs versus the target SU rate R_s for $\gamma_p = 5$ dB and $\gamma_s \in \{5, 10, 15\}$ dB. The proposed multi-SU admission achieves significantly more SUs than the conventional pairwise scheme, and the analytical step curves match the simulation results exactly in all three scenarios, validating the MAC-based characterization. Without power control, the selection of the strongest SUs yields a comparable or even larger average admitted set than the equal-gain case and produces fractional averages. For example, at $R_s = 1$ the no-power-control averages are 3.75, 5.47, and 7.19, versus 3, 5, and 7 under equal-gain, while at $R_s = 3$ it still admits 0.67, 1.03, and 1.92 SUs on average even when the equal-gain case has already dropped to 0 or 1 SU.

In Fig. 2, we plot the average per-SU rate $\bar{R}_s$ versus the PU rate R_p for $N = 3$ and $N = 5$ with $\gamma_p = 5$ dB and $\gamma_s = 10$ dB, comparing the MAC boundary, OMA, and the NOMA baseline. The RSMA points coincide with the MAC boundary and consistently exceed OMA and NOMA.

For instance, at $R_p = 1.0$ bit/s/Hz, RSMA achieves $\bar{R}_s = 1.36$ for $N = 3$ and 0.95 for $N = 5$, while OMA and NOMA yield 0.59 and 0.94 for $N = 3$, and 0.36 and 0.56 for $N = 5$. Moreover, increasing N reduces $\bar{R}_s$ due to rate sharing. When R_p approaches $C_p = 2.06$ the advantage of the MAC bound becomes more pronounced, where the MAC and RSMA values remain 1.01 for $N = 3$ and 0.74 for $N = 5$. In contrast, NOMA saturates at 0.59 and 0.35 and OMA drops to zero.

In Fig. 3, we plot the UAV rate R_s versus the minimum private-rate fraction α for $N = 4$ and $R_p = 0.5$ bit/s/Hz under $\gamma_p = 5$ dB, considering two cases $\gamma_s = 10$ dB and $\gamma_s = 5$ dB. We also compare the optimized SCA design, equal split, PU-margin, and the MAC upper bound. Also, the PU-margin uses $\sum_{i=1}^N \delta_i = \beta D_N^*$, where $C(\gamma_p, D_N^*) = R_p$, so that the PU is decoded with a specific rate margin while the residual power is left for the SU streams. It can be observed that the optimized design achieves the MAC upper bound when α is small. Meanwhile, R_s decreases as α increases since a stronger private-rate guarantee reduces the rate that can be supported on the common part, and lowering γ_s shifts all curves downward. For $\gamma_s = 10$ dB, the optimized scheme achieves $R_s = 1.24$ bit/s/Hz for $\alpha \leq 0.2$ and decreases to 0.46 at $\alpha = 0.8$, whereas for $\gamma_s = 5$ dB it drops from 0.85 at $\alpha = 0$ to 0.37 at $\alpha = 0.8$.

VI. CONCLUSION

This letter has presented an uplink cognitive access framework that allows multiple UAVs to share the spectrum of a licensed primary uplink while preserving the PU quality-of-service. A closed-form characterization of the maximum admissible number of UAVs was obtained from the MA region for equal UAV gains. Moreover, we also consider an RSMA optimization to ensure a minimum private-rate ratio while maximizing the achievable SU rate. Numerical results have demonstrated that the proposed framework can admit significantly more UAVs than conventional spectrum sharing while maintaining the PU target rate. In addition, optimization result converges stably and attains SU rates close to the MAC upper bound with moderate private-rate.

APPENDIX A PROOF OF LEMMA 1

Let $u(x) = \log_2(1 + \gamma_p + x\gamma_s)$, so that

$$f(x) = \frac{u(x) - R_p}{x}, \quad x > 0. \quad (18)$$

The derivative of $u(x)$ is

$$u'(x) = \frac{\gamma_s}{(1 + \gamma_p + x\gamma_s) \ln 2} = \frac{\gamma_s \log_2 e}{1 + \gamma_p + x\gamma_s}. \quad (19)$$

By differentiating $f(x)$ we obtain

$$f'(x) = \frac{xu'(x) - (u(x) - R_p)}{x^2} = \frac{z(x)}{x^2}, \quad (20)$$

where $z(x) = xu'(x) - u(x) + R_p$. Hence the sign of $f'(x)$ is the same as the sign of $z(x)$.

Using the expression of $u'(x)$, a simple calculation gives

$$z'(x) = -\frac{x\gamma_s^2 \log_2 e}{(1 + \gamma_p + x\gamma_s)^2} < 0, \quad x > 0. \quad (21)$$

Thus $z(x)$ is strictly decreasing on $(0, \infty)$. Moreover,

$$\lim_{x \rightarrow 0^+} z(x) = 0 - \log_2(1 + \gamma_p) + R_p = R_p - \log_2(1 + \gamma_p). \quad (22)$$

Under (2), we have $R_p \leq \log_2(1 + \gamma_p)$, and therefore $\lim_{x \rightarrow 0^+} z(x) \leq 0$. Since $z(x)$ is strictly decreasing, $z(x) < 0$ for all $x > 0$, which implies $f'(x) = \frac{z(x)}{x^2} < 0$. Hence $f(x)$ is strictly decreasing on $(0, \infty)$, which proves Lemma 1.

APPENDIX B PROOF OF PROPOSITION 1

From (3) and (4), the constraints for subsets containing the PU and any m UAVs are equivalent to $R_s \leq f(m)$, with $m \in \{1, \dots, N\}$. By Lemma 1, $f(x)$ is strictly decreasing on $(0, \infty)$. For the given $R_s > 0$, let x^* and $N_{\max} = \lfloor x^* \rfloor$.

If $m \leq N_{\max} \leq x^*$, then $f(m) \geq f(x^*) = R_s$, so all MAC constraints are satisfied for every $m \leq N_{\max}$. If $m \geq N_{\max} + 1 > x^*$, then $f(m) < f(x^*) = R_s$, hence the corresponding MAC constraint is violated. Therefore, no $(N_{\max} + 1)$ -UAV configuration is feasible, and $N_{\max}$ is the largest admissible number of UAVs.

REFERENCES

- [1] B. Li, Z. Fei, and Y. Zhang, "UAV Communications for 5G and Beyond: Recent Advances and Future Trends," *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 2241–2263, Apr. 2019.
- [2] H. Liu, Z. Bai, H. Lei, G. Pan, K. J. Kim, and T. A. Tsiftsis, "A Rate Splitting Strategy for Uplink CR-NOMA Systems," arXiv preprint, Oct. 2021.
- [3] Y. Mao, O. Dizdar, B. Clerckx, R. Schober, P. Popovski, and H. V. Poor, "Rate-Splitting Multiple Access: Fundamentals, Survey, and Future Research Trends," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2073–2126, 2022.
- [4] W. Liang, Z. Ding, Y. Li, and L. Song, "User Pairing for Downlink NOMA Networks Using Matching Algorithm," *IEEE Transactions on Communications*, vol. 65, no. 12, pp. 5319–5332, Dec. 2017.
- [5] Y. Huang, X. He, and Y. Du, "Research on User Pairing Techniques Based on NOMA in Cognitive Radio Networks," *Journal of Computer and Communications*, vol. 11, no. 3, pp. 43–55, Mar. 2023.
- [6] S. Khisa, M. Elhattab, C. Assi, and S. Sharafeddine, "Optimizing Multi-User Uplink Cooperative Rate-Splitting Multiple Access: Efficient User Pairing and Resource Allocation with Gradient-Based Meta Learning," *IEEE Transactions on Communications*, vol. 73, no. 9, pp. 7366–7380, 2025.
- [7] P. T. Q. Trang, H. X. Son, B. N. Son, D. T. Duong, and T. A. Vu, "A Novel Rate Splitting Scheme for Uplink Cognitive Radio Networks," *IEEE Wireless Communications Letters*, vol. 1, no. 1, p. 99, 2025.
- [8] D. Tse and P. Viswanath, *Fundamentals of wireless communication*. Cambridge: Cambridge University Press, 2005, 10715.
- [9] "CVX: matlab software for disciplined convex programming," Sep. 2013.